ECE 241
Digital Design



Dr. D. Capson, P.Eng.

Department of Electrical & Computer Engineering

University of Victoria

capson@uvic.ca

©2026 Dr. D. Capson. All rights reserved.

on-line course materials

All lab information, lecture power points, practice problems, solutions, grades, announcements will be available on our course ***Brightspace*** page

©2026 Dr. D. Capson. All rights reserved.

Course Administration

Office hours: by appointment (in person or Zoom)

Lectures: Tuesdays, Wednesdays, Fridays
10:30am - 11:20am, ~~CORA221~~ **BWC A104**

Labs: ALL Labs are in ELW A359. Sections and schedule will be confirmed soon.
(you must attend all labs and complete all lab requirements to pass the course)

Grading:

Labs	30%	dates: consult your timetable
Midterm #1	15%	Friday, Oct 9, 2026
Midterm #2	15%	Wednesday, Nov 4, 2026
Final Exam	40%	date TBD
		(you must pass exam to pass the course)

Note: All Labs, Midterms, and Final Exam will be face-to-face

©2026 Dr. D. Capson. All rights reserved.

Please read the official ECE241 Course Outline

available on the ECE241 course *Brightspace* page

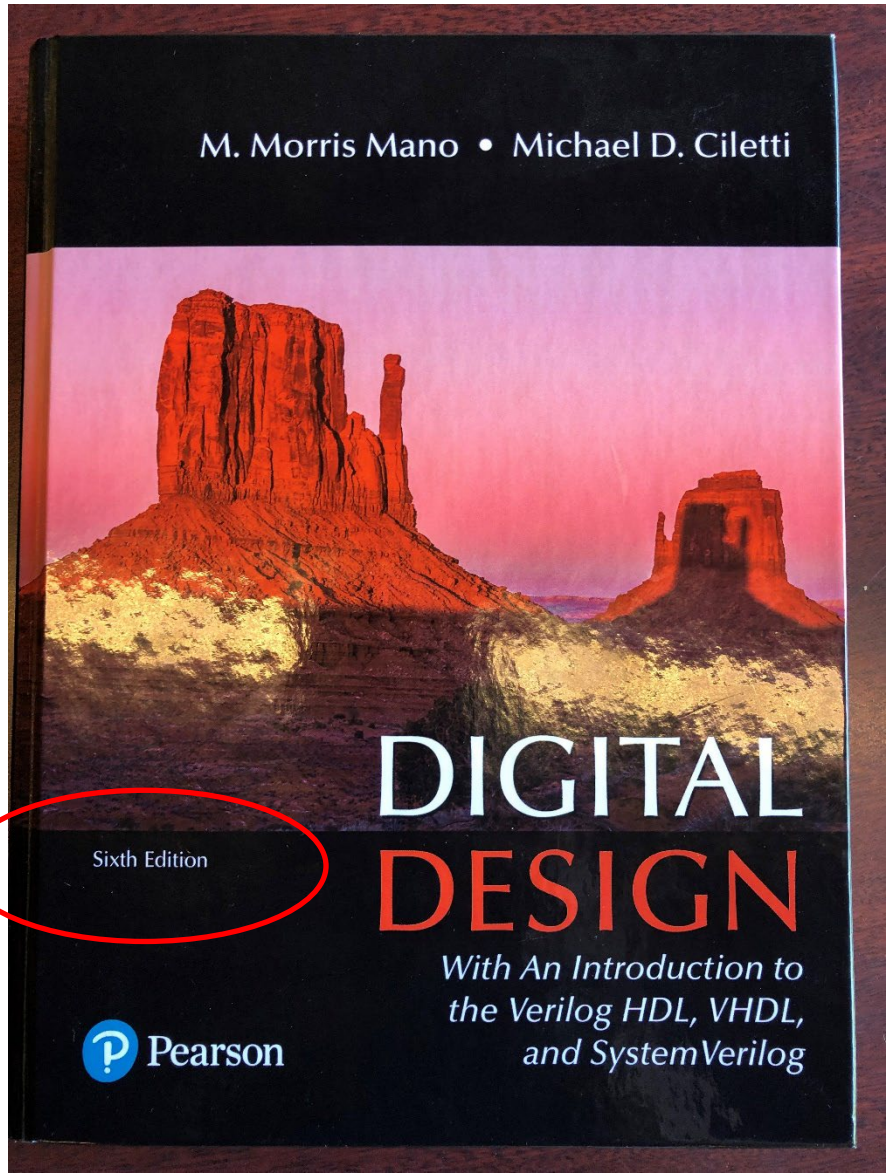
©2026 Dr. D. Capson. All rights reserved.

Course policies on use of AI

- Use of ChatGPT (or other AI tools) is permitted.
- Do NOT submit any AI-generated materials as your own work. This may be considered as plagiarism which is a serious academic offence.

Course Textbook

Note:
6th edition



©2026 Dr. D. Capson. All rights reserved.

How to do well in this course

- Text book will be followed closely
- Stay connected with the course web site – check in frequently for latest info
- Labs are critical to understanding the material
- Keep up with regular reading

ENJOY the material !



©2026 Dr. D. Capson. All rights reserved.

Course Topics

- Binary/Hex representations, conversions and arithmetic
- Boolean algebra
- Logic gates and their electronic characteristics
- Combinational circuits including multiplexers, decoders, adders, multipliers, ALU
- Sequential circuits including latches, flip-flops, registers, counters and memory systems
- Finite state machine analysis and design
- Introduction to Hardware Description Language (VHDL)
- Programmable logic devices including PLA, CPLD, FPGA
- Introduction to computer organization

©2026 Dr. D. Capson. All rights reserved.

Why is this course important ?

The computer plays an increasing and central role in everyday life and costs have decreased dramatically. Computers are "embedded" nearly everywhere…

Sales of microprocessor chips for embedded applications are estimated to be at least *10,000x* the number for use in laptops/workstations.

©2026 Dr. D. Capson. All rights reserved.

Why is this course important ?

For example, how many computers can be found in a recent-model automobile?

©2026 Dr. D. Capson. All rights reserved.

Why is this course important ?

For example, how many computers can be found in a recent-model automobile?

typically 40-100

… and increasing

©2026 Dr. D. Capson. All rights reserved.

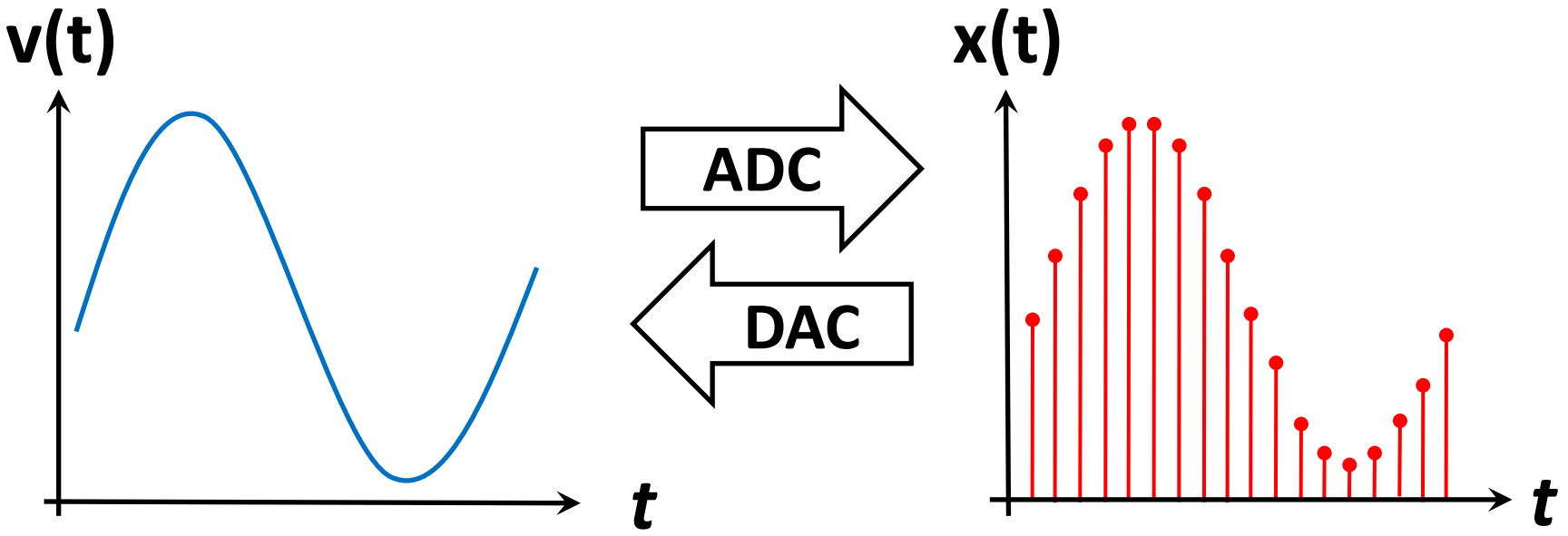
Why is this course important ?

AND…

Digital Signal Processing (DSP)
has had a profound impact on society
and virtually every area of
Electrical & Computer Engineering

©2026 Dr. D. Capson. All rights reserved.

What is Digital Signal Processing?



Analog signal

Vary continuously over time
Occur naturally in the world
Susceptible to noise corruption and degradation

Digital samples

Assume discrete values
Easy to store, repeatable
Many DSP algorithms.
e.g. compression

©2026 Dr. D. Capson. All rights reserved.

Some history …

George Boole – English mathematician. In 1854 demonstrates the mathematics of logic. Sets the foundations of *Boolean algebra* which we will use frequently in this course.

Claude Shannon – first to note similarities between Boolean algebra and electronic telephone switches. His 1940 MIT Master's thesis is credited for introducing binary digital electronics.

©2026 Dr. D. Capson. All rights reserved.

Gordon E. Moore – co-founder of Intel. Predicted home computers and the smart phone in his famous 1965 article:

"Integrated circuits will lead to such wonders as home computers - or at least terminals connected to a central computer - automatic controls for automobiles, and personal portable communications equipment."

Moore's original paper is available from:

Electronics, Volume 38, Number 8, April 19, 1965

or find the reprint on IEEE Xplore:

IEEE SSCS Newsletter September 2006

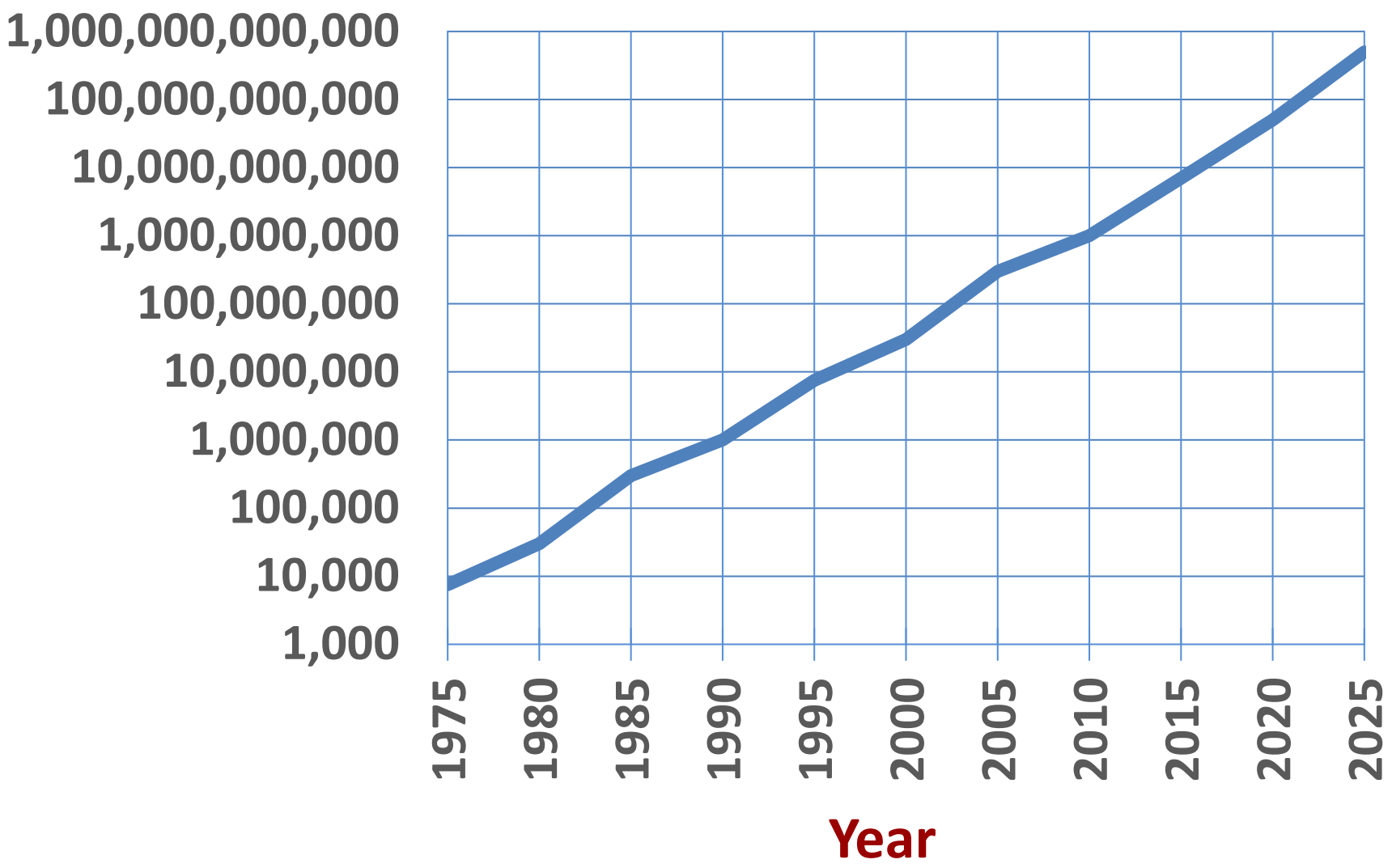
©2026 Dr. D. Capson. All rights reserved.

Moore's Law – his 1965 article also predicted a doubling of the number of transistors that could be put on a chip every 18-24 months

Proved to be accurate for upwards of 61 years (so far) and has *fundamentally* impacted the speed, cost, power consumption (i.e. HEAT) and physical size of processors, memory, transducers, pixels, etc.

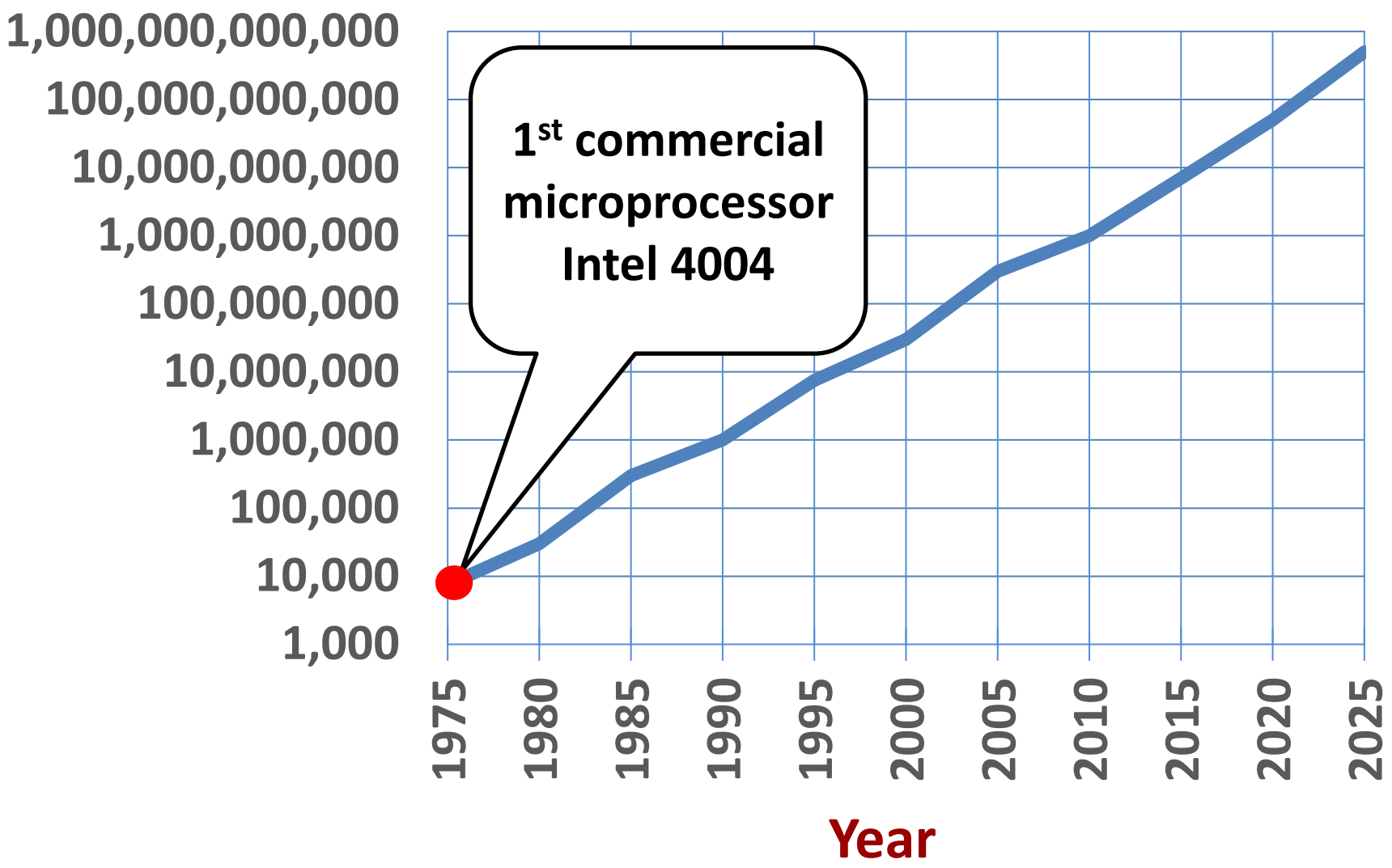
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



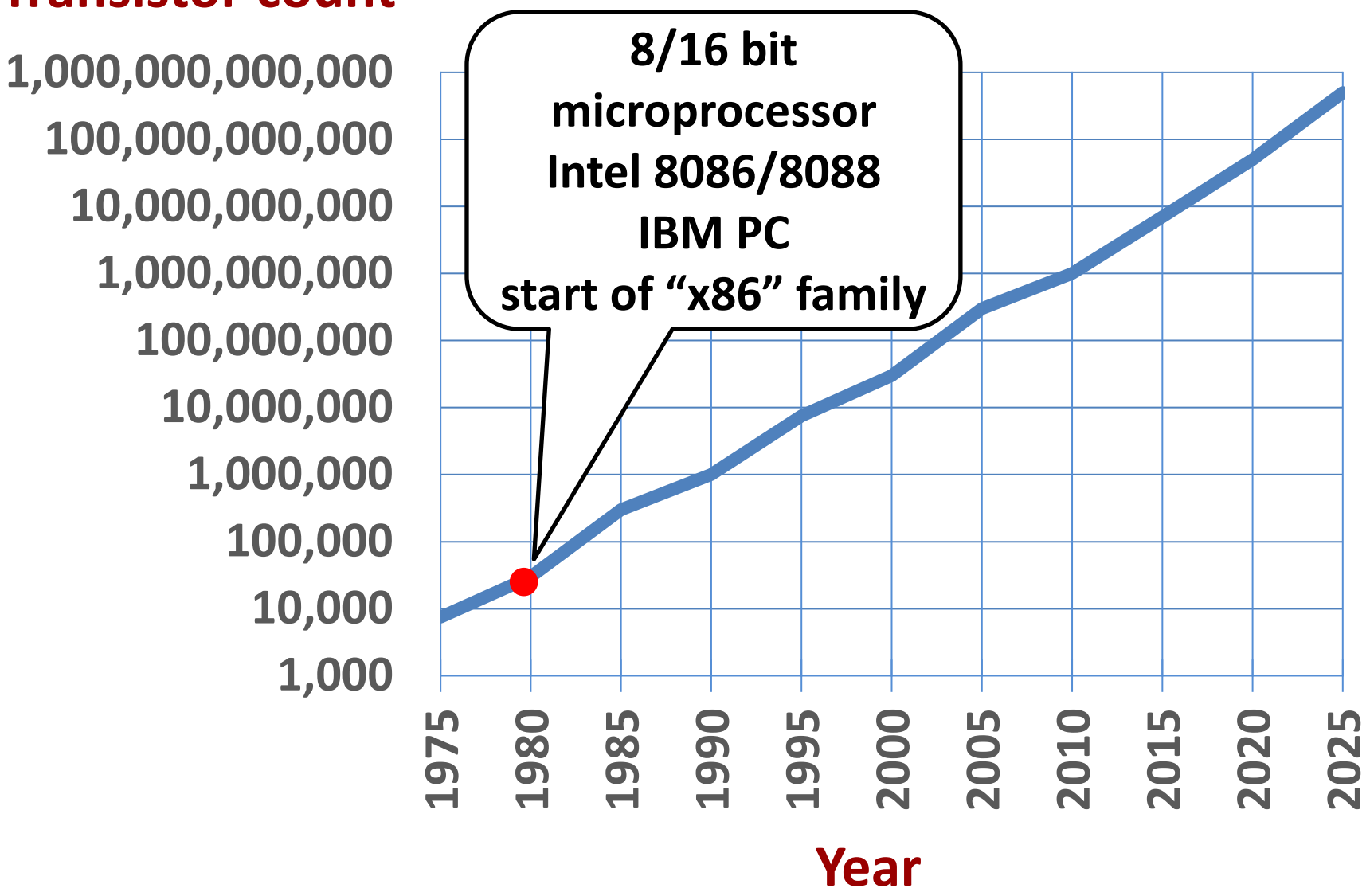
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



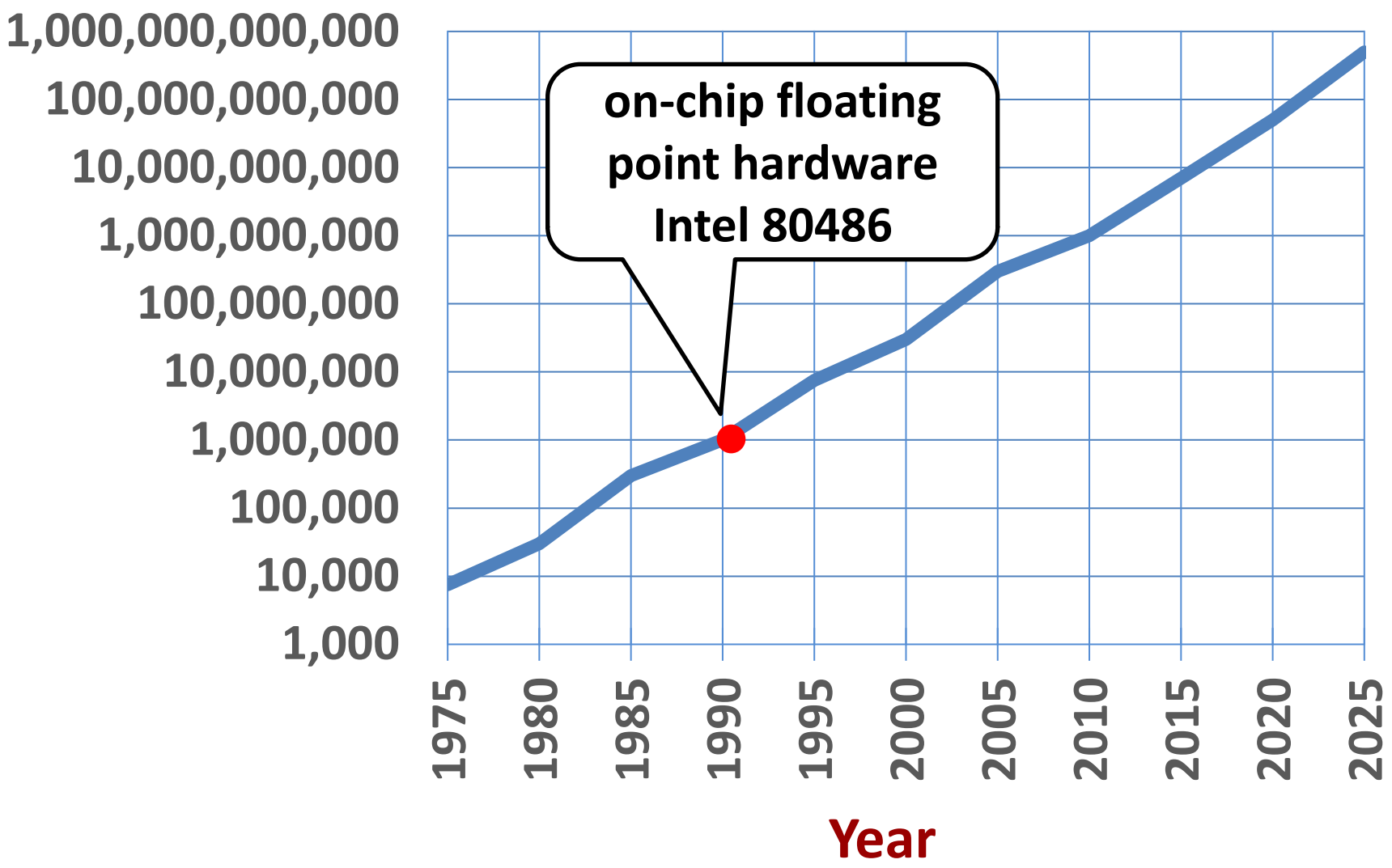
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



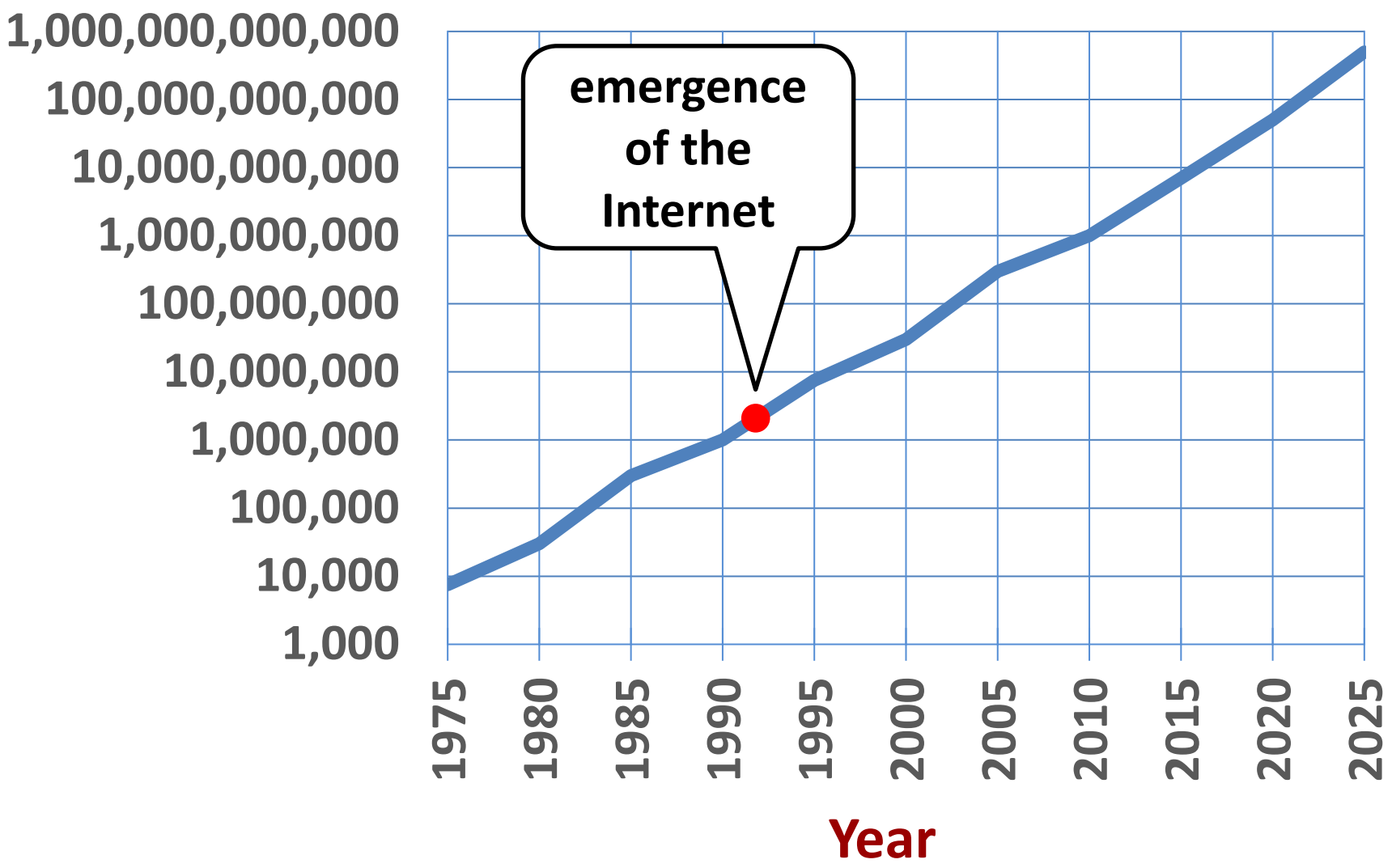
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



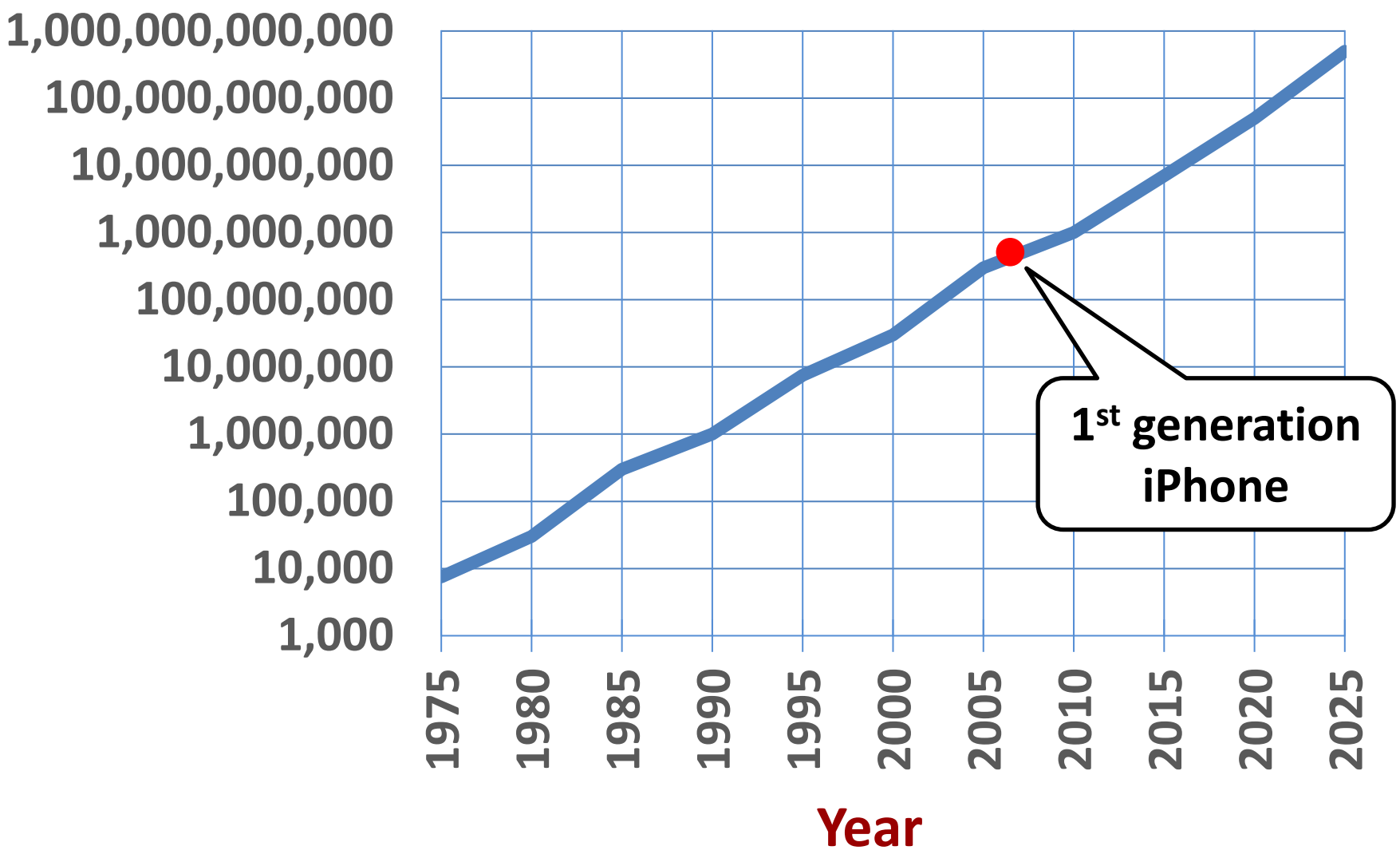
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



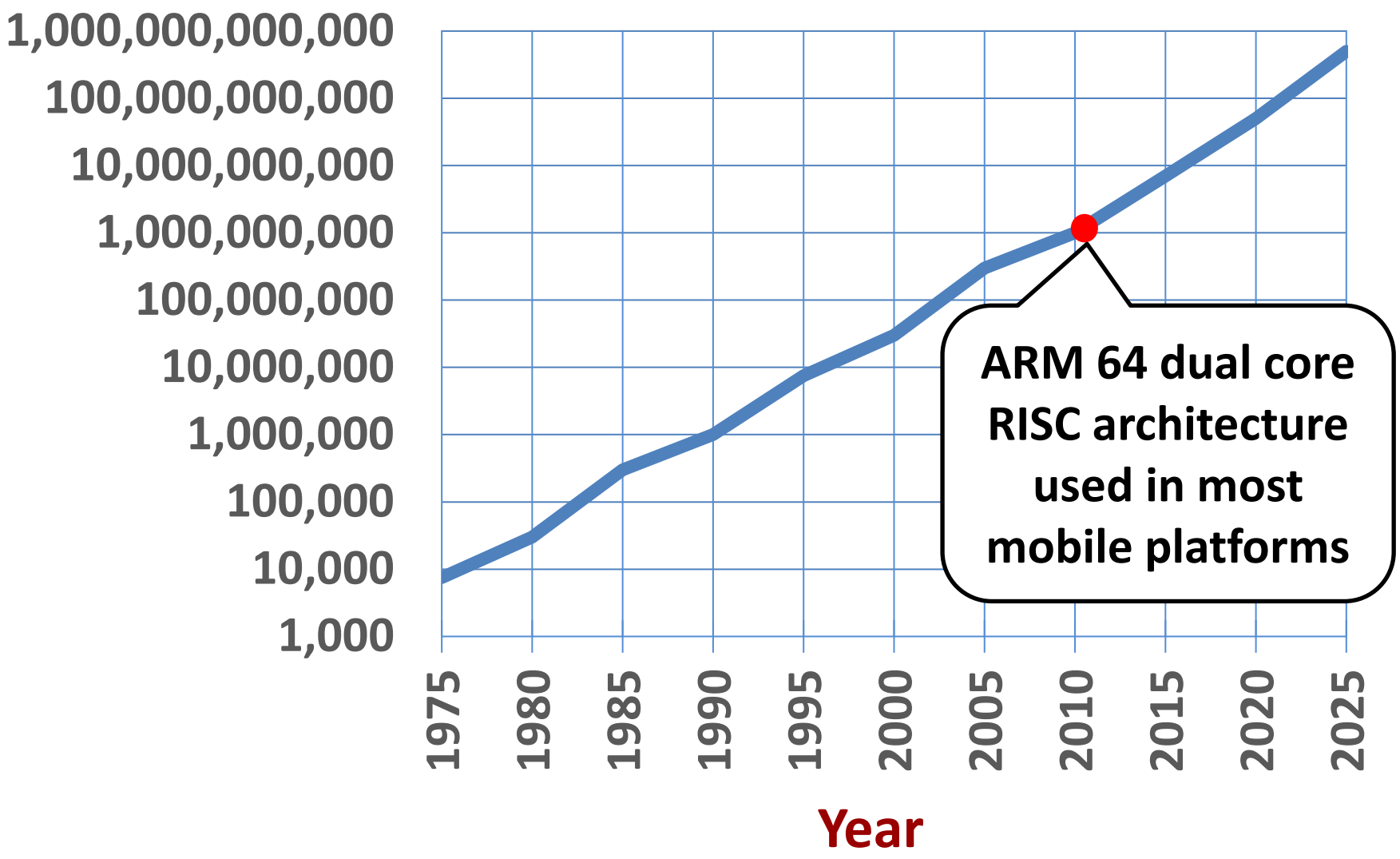
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



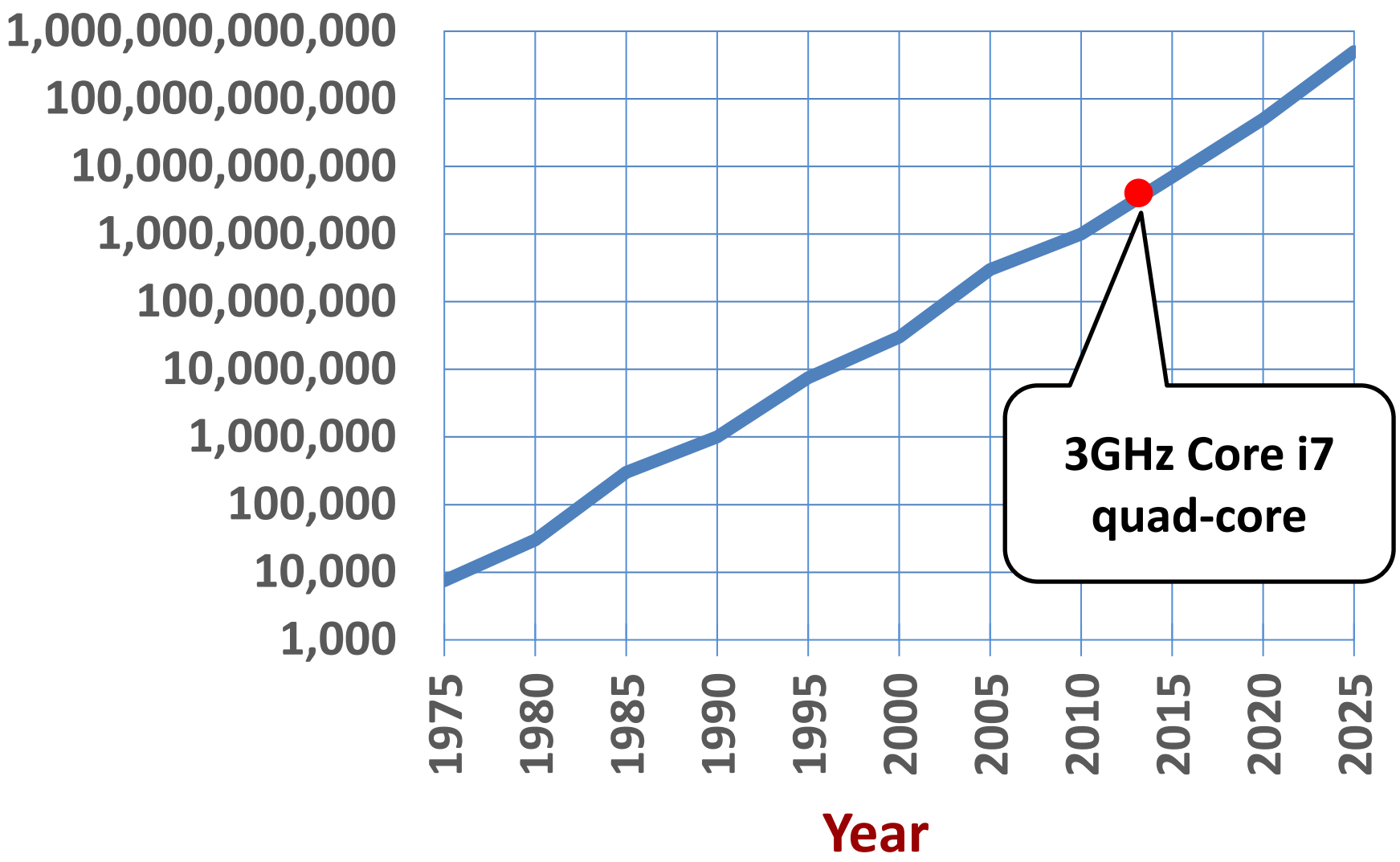
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



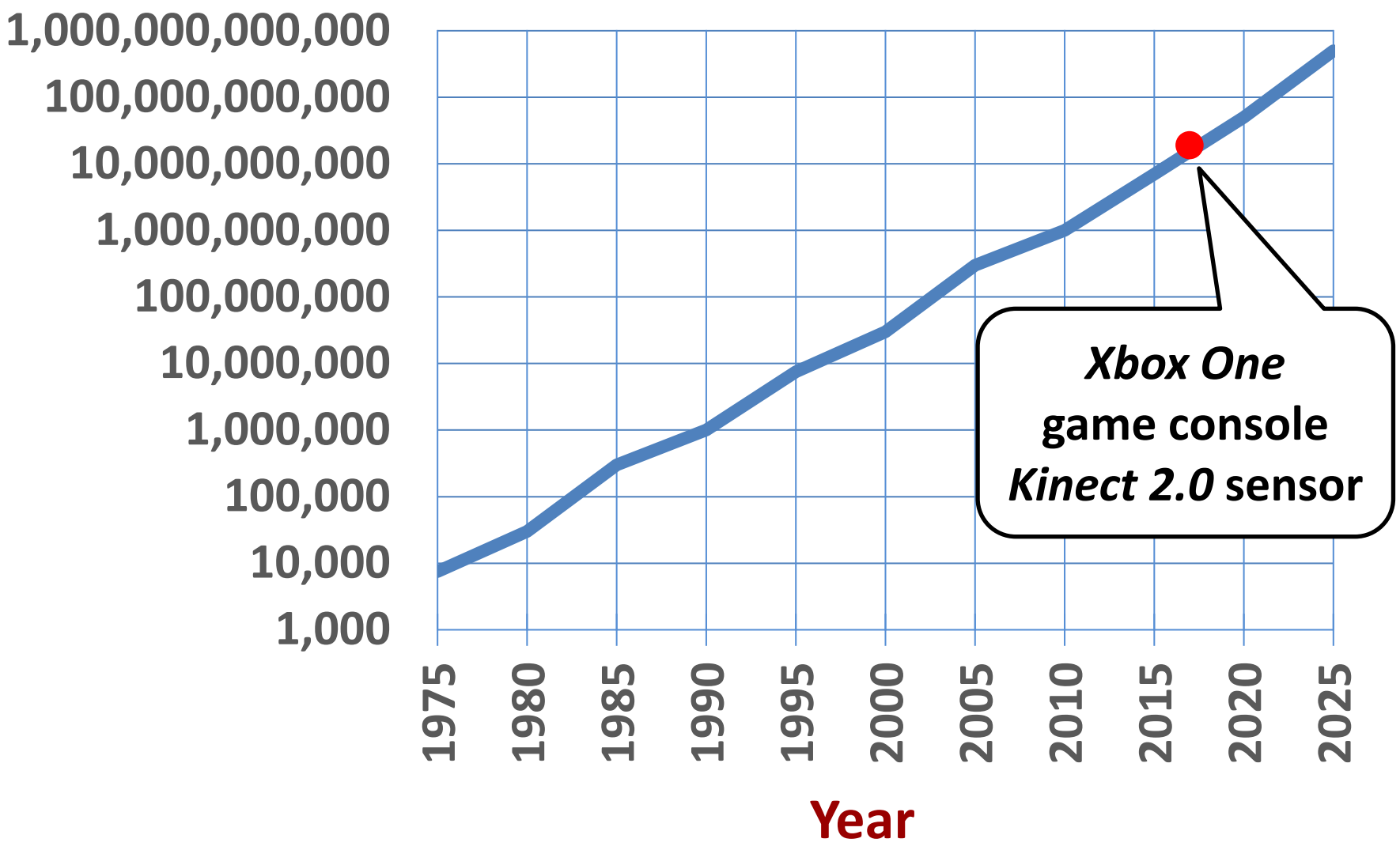
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



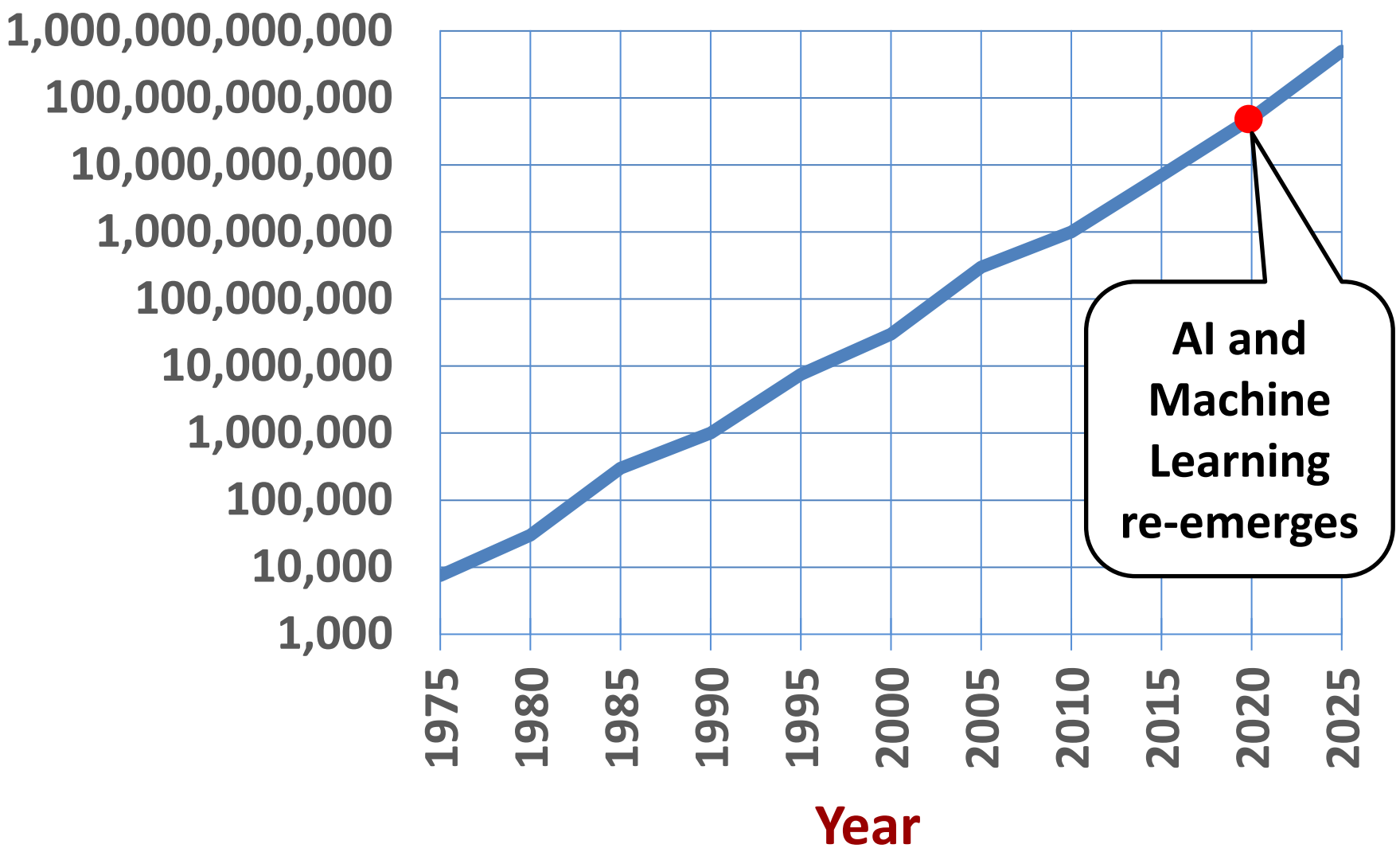
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



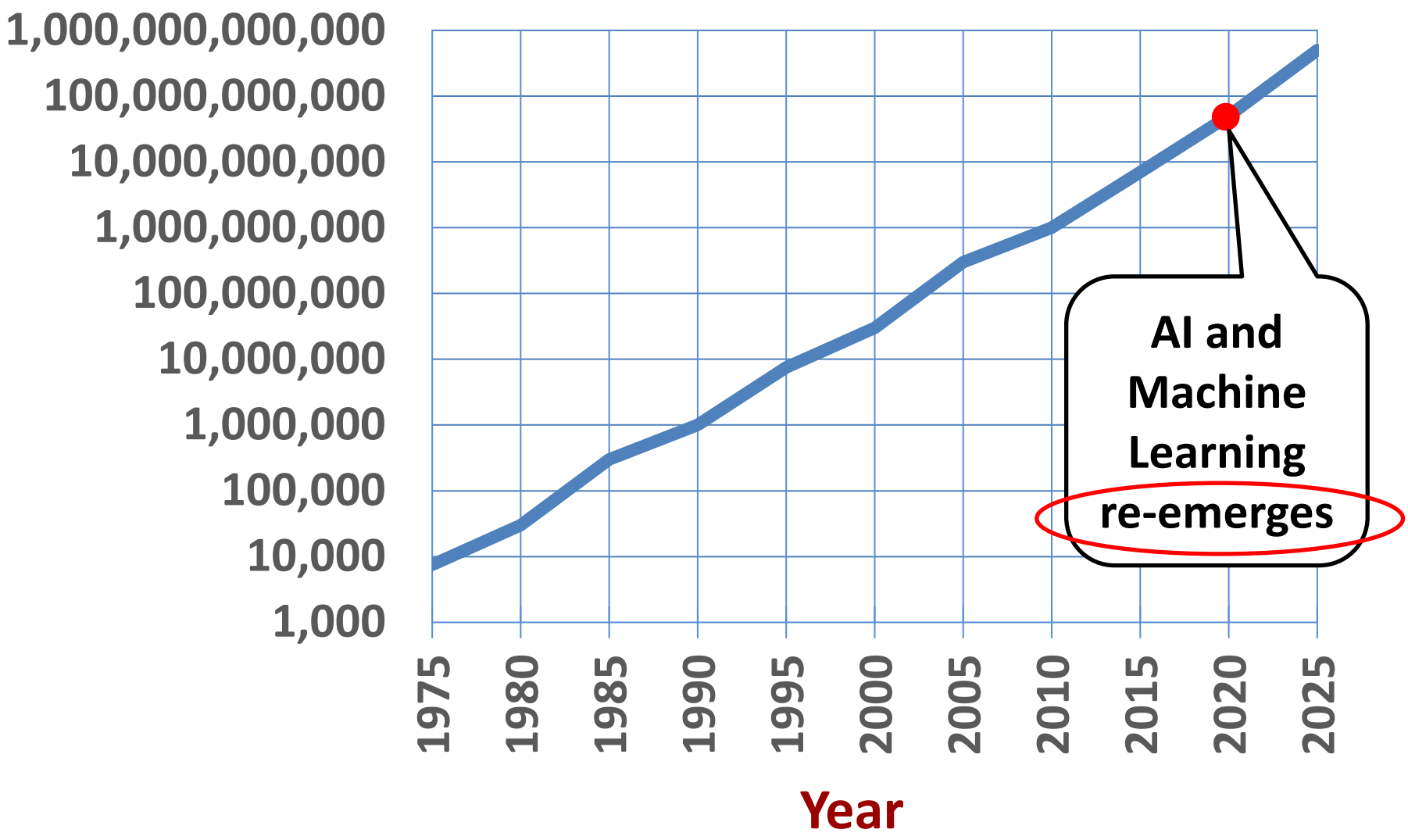
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



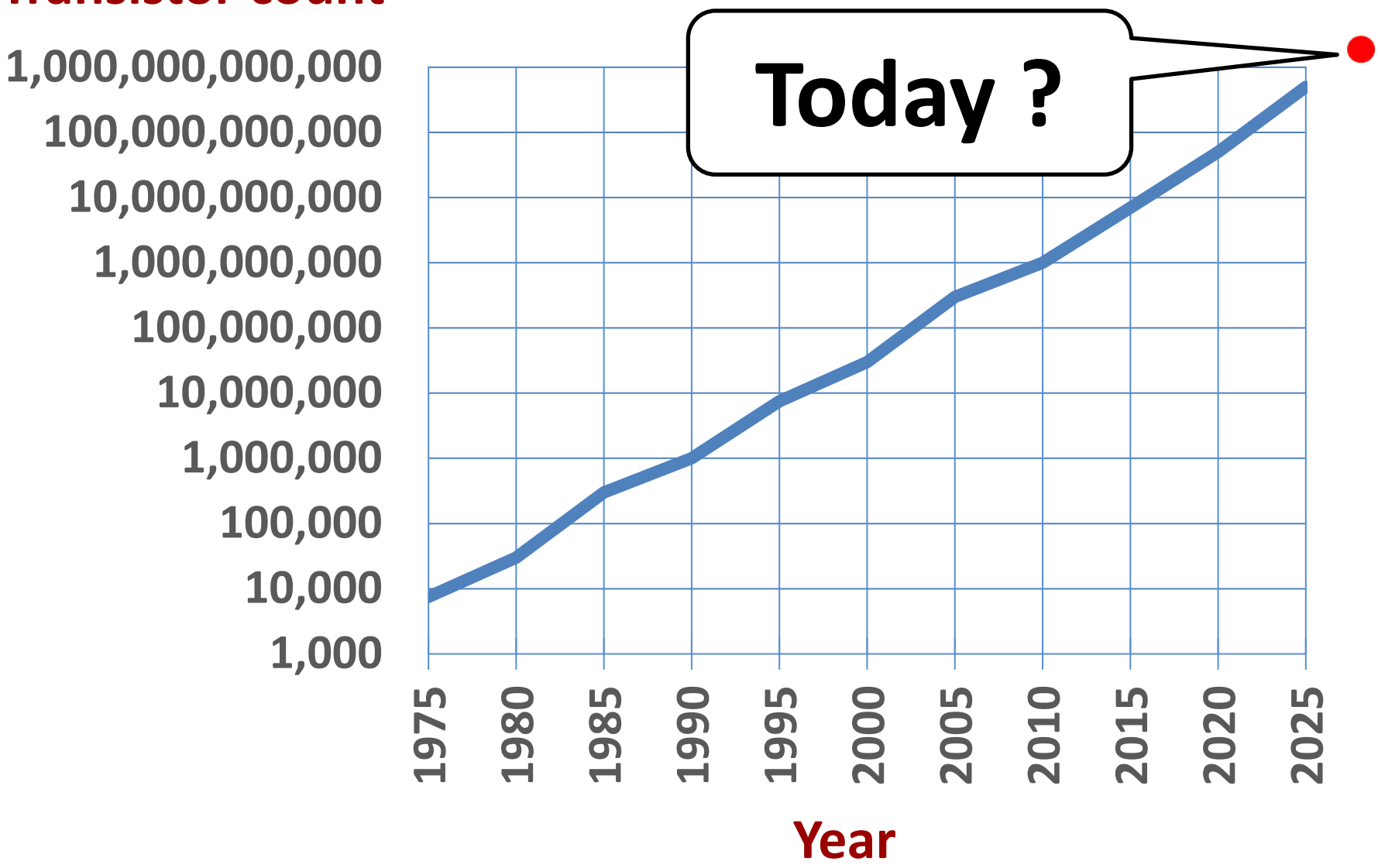
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



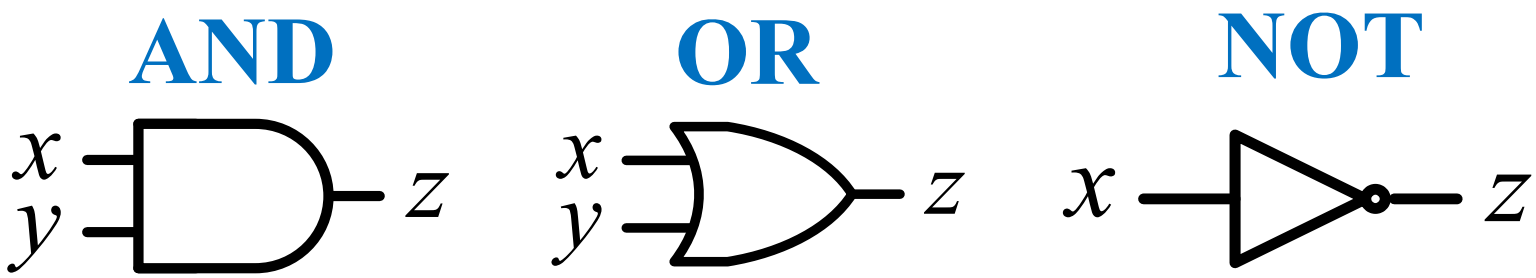
©2026 Dr. D. Capson. All rights reserved.

Moore's Law



©2026 Dr. D. Capson. All rights reserved.

It requires 2 to 6 transistors to make 1 logic gate.



Designing circuits with logic gates is the main focus of this course !

©2026 Dr. D. Capson. All rights reserved.

How are logic gates built?

Implementation of logic gates on IC's is based on CMOS transistor technology:

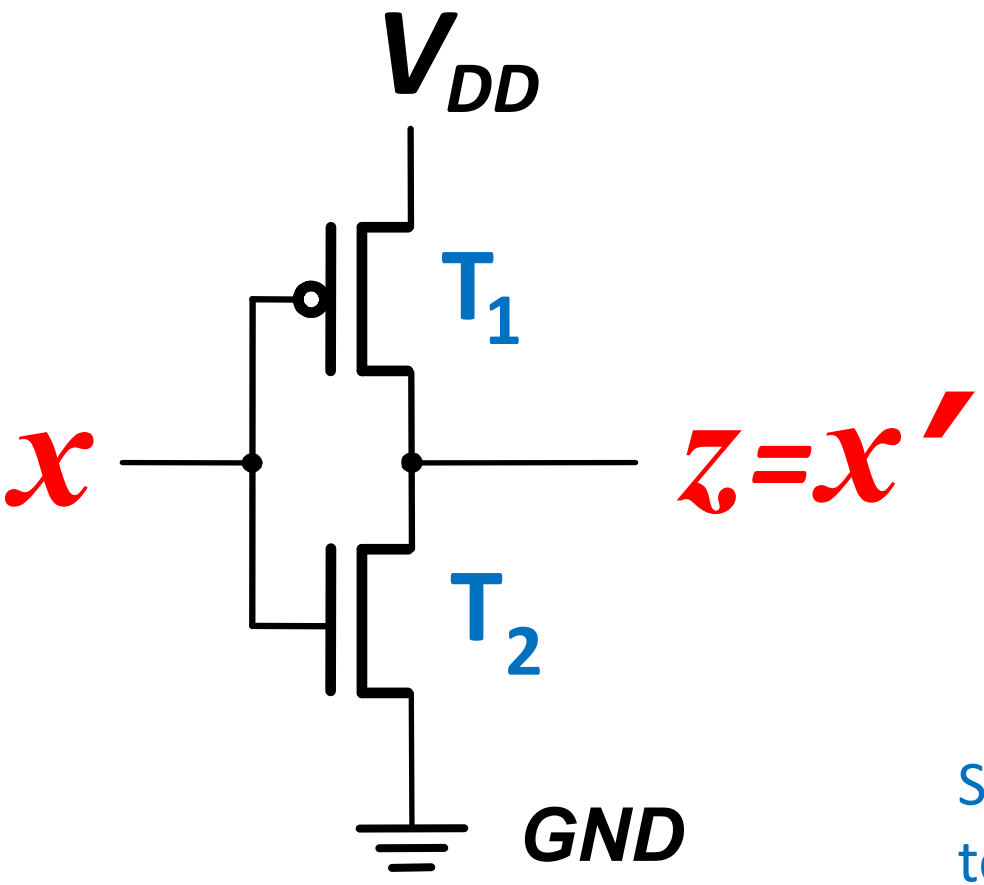
- very low power consumption
- good noise immunity
- high density (i.e. small size!)

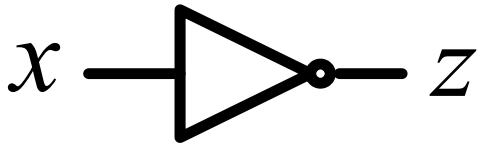
CMOS = complementary metal oxide semi-conductor

©2026 Dr. D. Capson. All rights reserved.

How are logic gates built?

Example: NOT gate using CMOS





x	T_1	T_2	z
0	on	off	1
1	off	on	0

See **Section A.1** of your textbook for some examples

©2026 Dr. D. Capson. All rights reserved.

Logic gates work with "digital" inputs and outputs …

In digital circuits, inputs (e.g. x, y) and outputs (e.g. z) are interpreted as *discrete* values. Two-level, or "*binary*" values may be described as:

0 and 1

False (F) and True (T)

Low (L) and High (H)

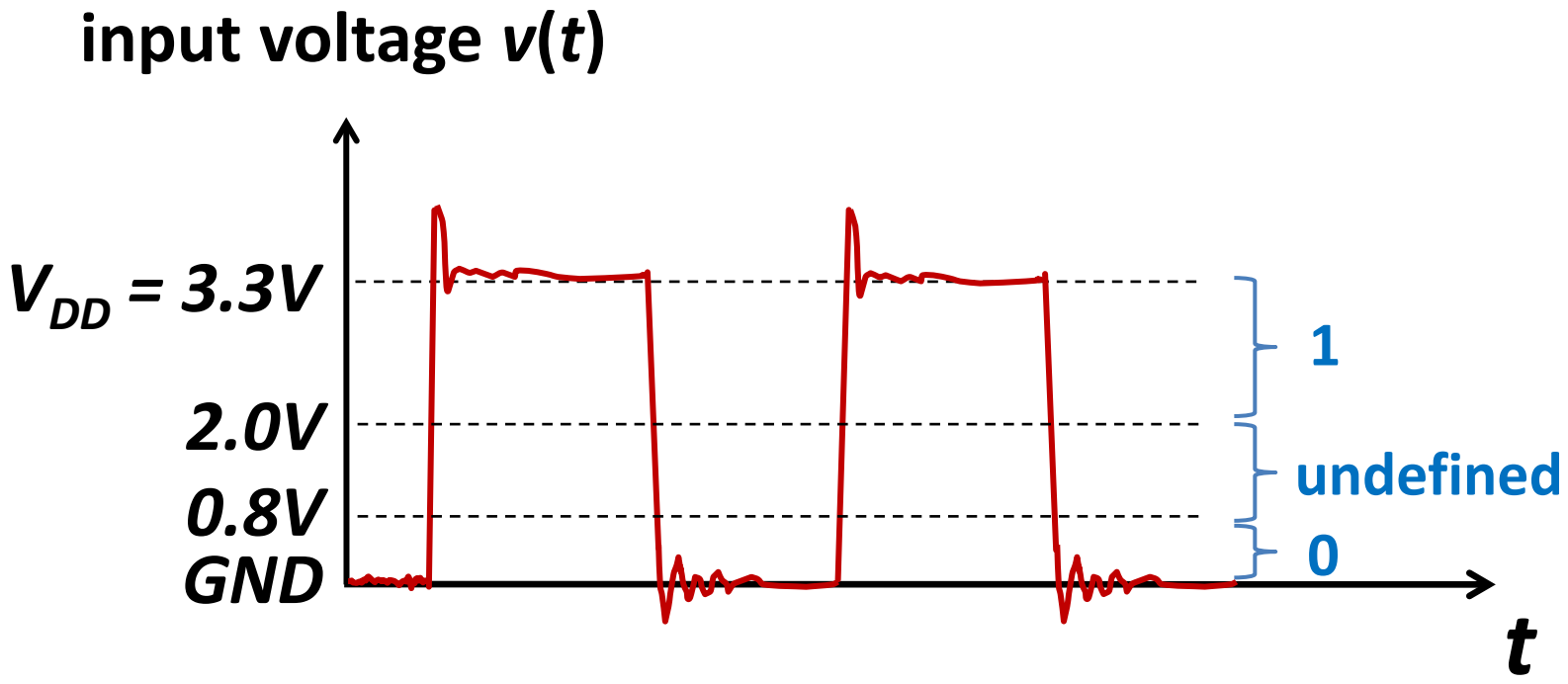
OFF and ON

Open and Closed

But real digital circuits use voltage ranges …

©2026 Dr. D. Capson. All rights reserved.

What do real "digital" signals look like?



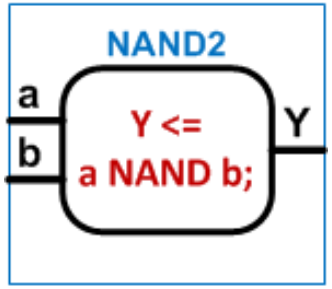
Notes:

(1) voltage ranges are different for inputs and outputs to provide noise "margin"

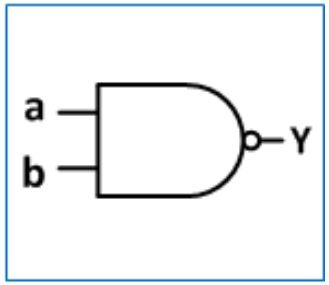
(2) other common values used for V_{DD} are 5V, 2.8V, 1.8V

©2026 Dr. D. Capson. All rights reserved.

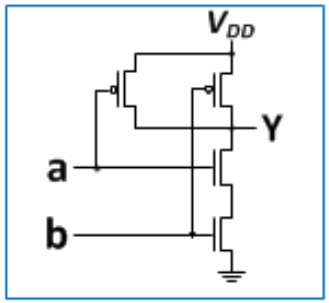
Layers of design abstraction



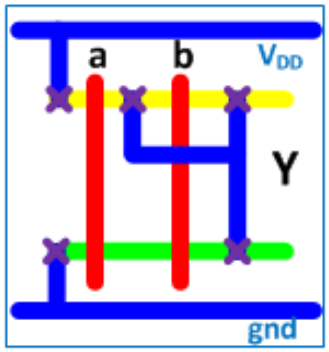
behavioral descriptions



logic gates



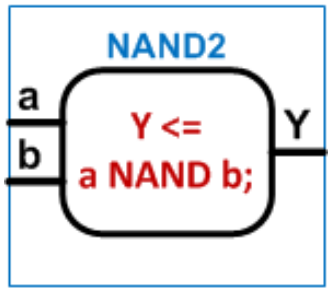
transistors



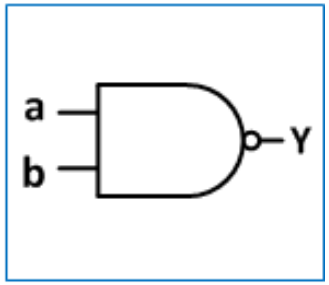
IC layout

©2026 Dr. D. Capson. All rights reserved.

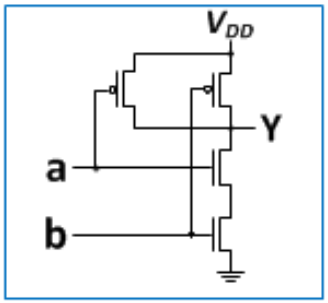
ECE241 will work mostly in these 2 layers



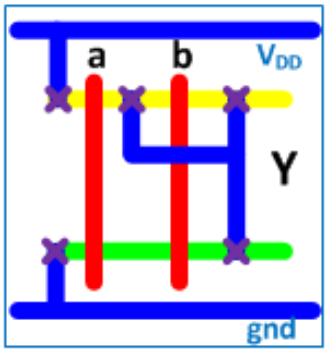
behavioral descriptions



logic gates



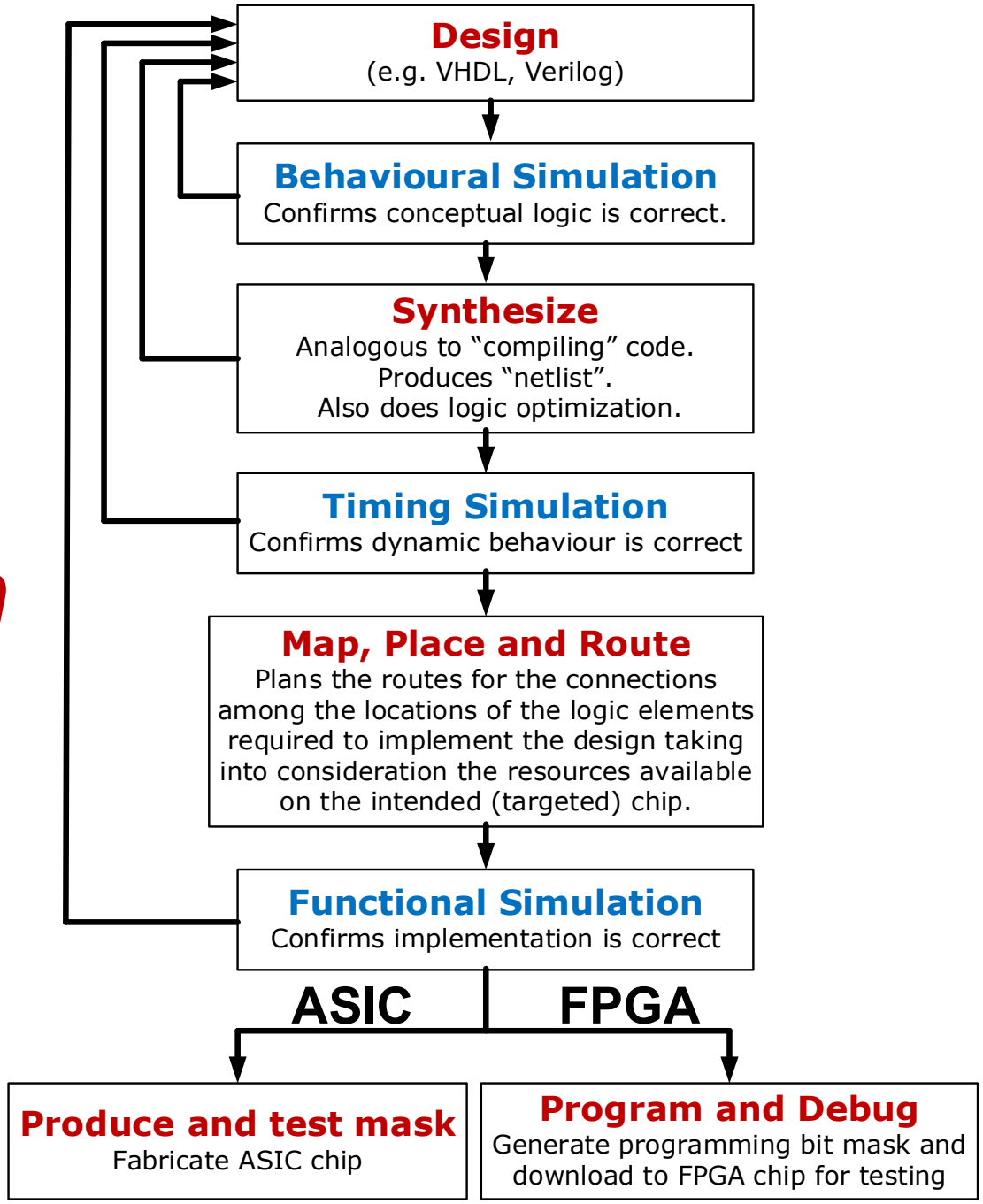
transistors



IC layout

©2026 Dr. D. Capson. All rights reserved.

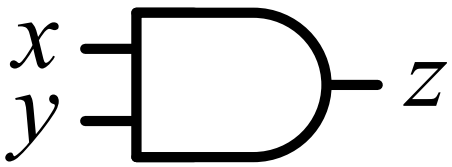
Modern *Electronic Design Automation* (EDA) Flow



©2026 Dr. D. Capson. All rights reserved.

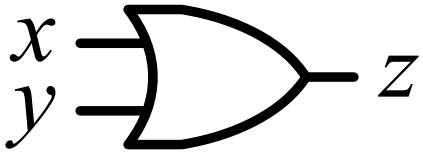
Truth tables …

AND



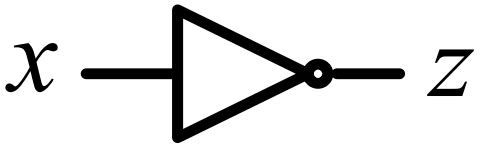
x	y	z
F	F	F
F	T	F
T	F	F
T	T	T

OR



x	y	z
F	F	F
F	T	T
T	F	T
T	T	T

NOT

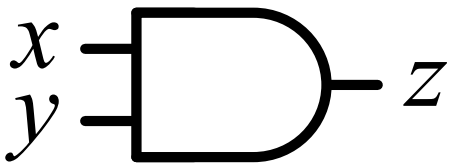


x	z
F	T
T	F

©2026 Dr. D. Capson. All rights reserved.

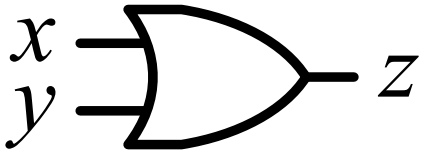
Most of the time we will use T==1. This is called "*positive*" logic …

AND



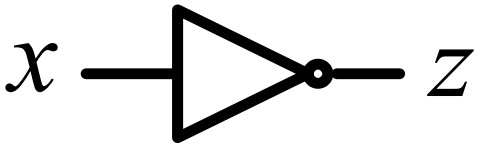
x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

OR



x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

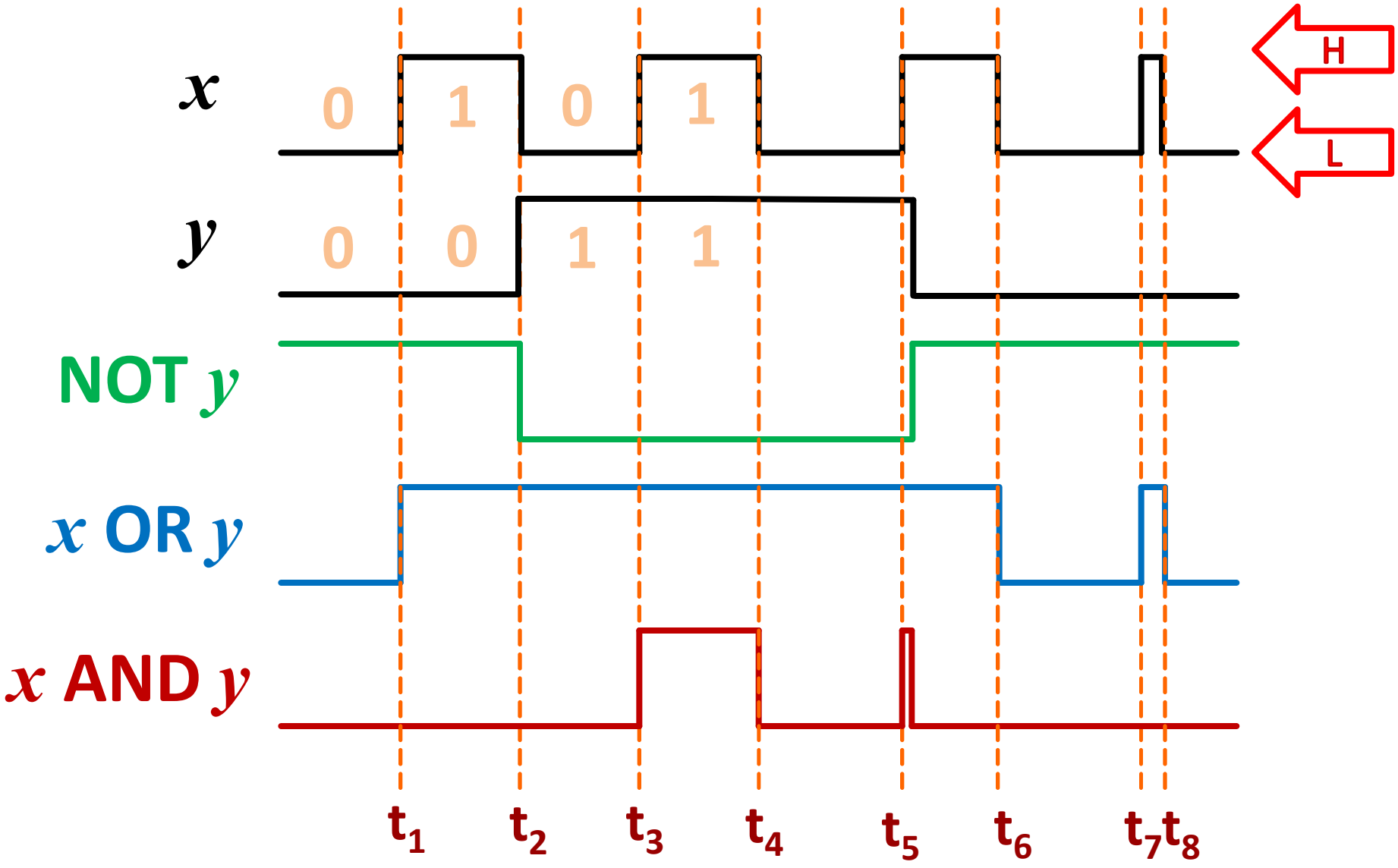
NOT



x	z
0	1
1	0

©2026 Dr. D. Capson. All rights reserved.

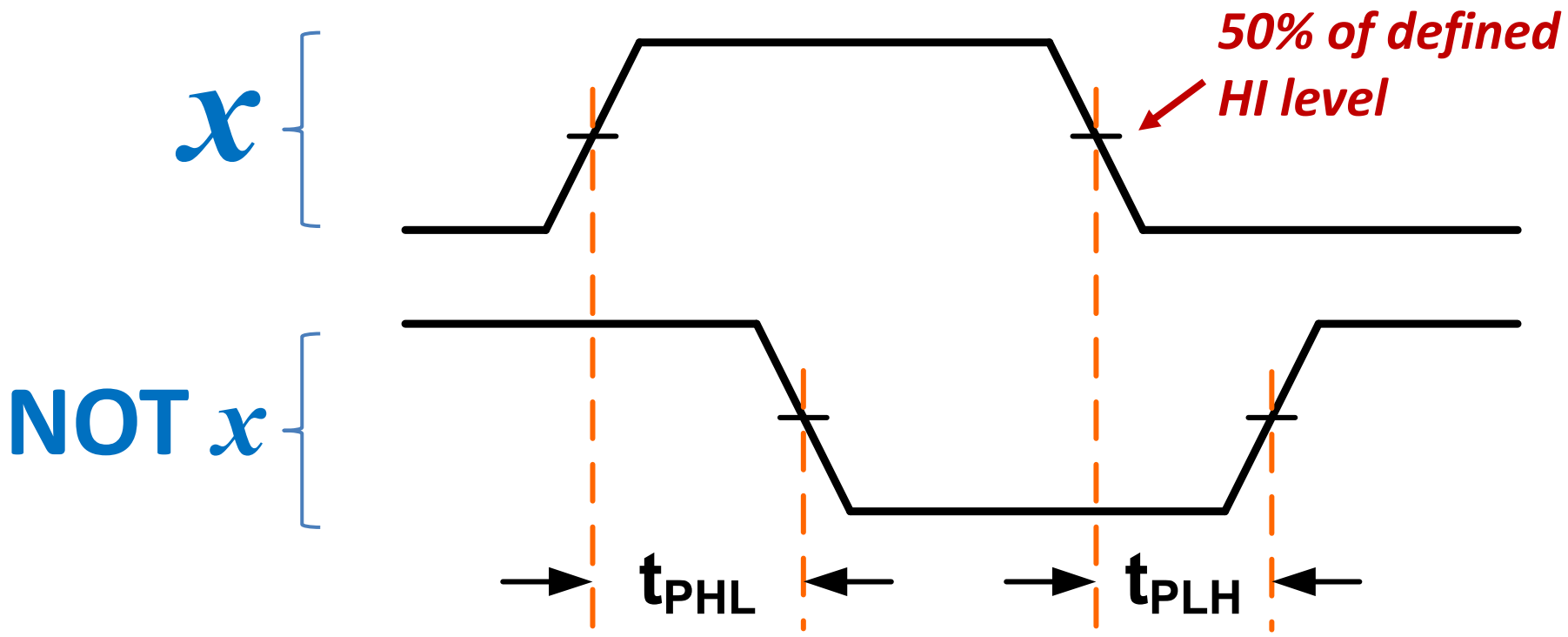
Timing example - idealized edges depicted



©2026 Dr. D. Capson. All rights reserved.

How do real logic gates behave at edges?

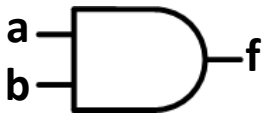
Example: NOT gate



Note: (1) propagation delays t_{PHL} and t_{PLH} are usually NOT exactly equal
(2) typical values for t_{PHL} = 23ps, t_{PLH} = 28ps (for CMOS)

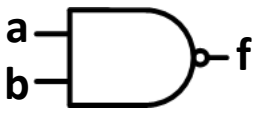
©2026 Dr. D. Capson. All rights reserved.

Basic 2-input Logic Gates



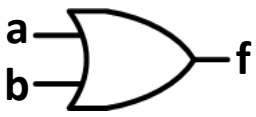
a	b	f
0	0	0
0	1	0
1	0	0
1	1	1

AND



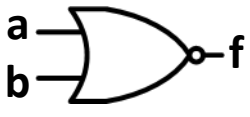
a	b	f
0	0	1
0	1	1
1	0	1
1	1	0

NAND



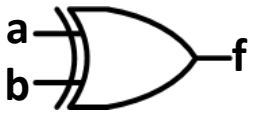
a	b	f
0	0	0
0	1	1
1	0	1
1	1	1

OR



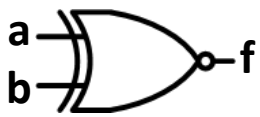
a	b	f
0	0	1
0	1	0
1	0	0
1	1	0

NOR



a	b	f
0	0	0
0	1	1
1	0	1
1	1	0

XOR

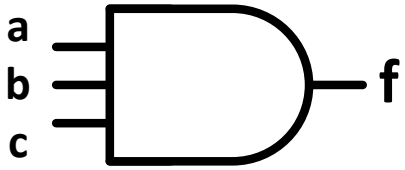


a	b	f
0	0	1
0	1	0
1	0	0
1	1	1

XNOR

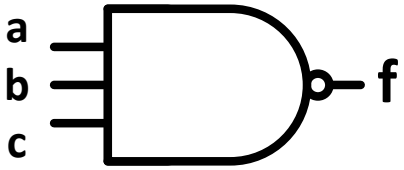
©2026 Dr. D. Capson. All rights reserved.

Examples of 3-input Logic Gates



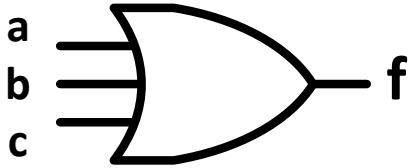
a	b	c	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

AND



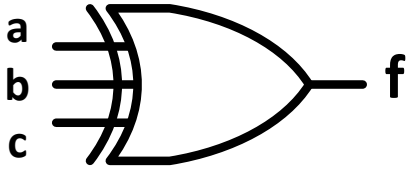
a	b	c	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

NAND



a	b	c	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

OR

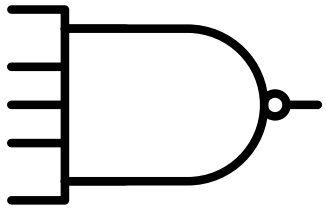


a	b	c	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

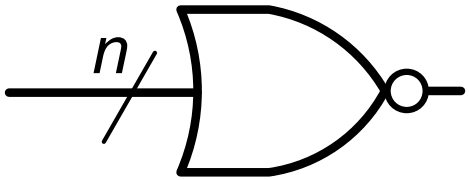
XOR

©2026 Dr. D. Capson. All rights reserved.

Generalize to *n*-input Logic Gates …



NAND



NOR

Logically, this can be extended indefinitely.

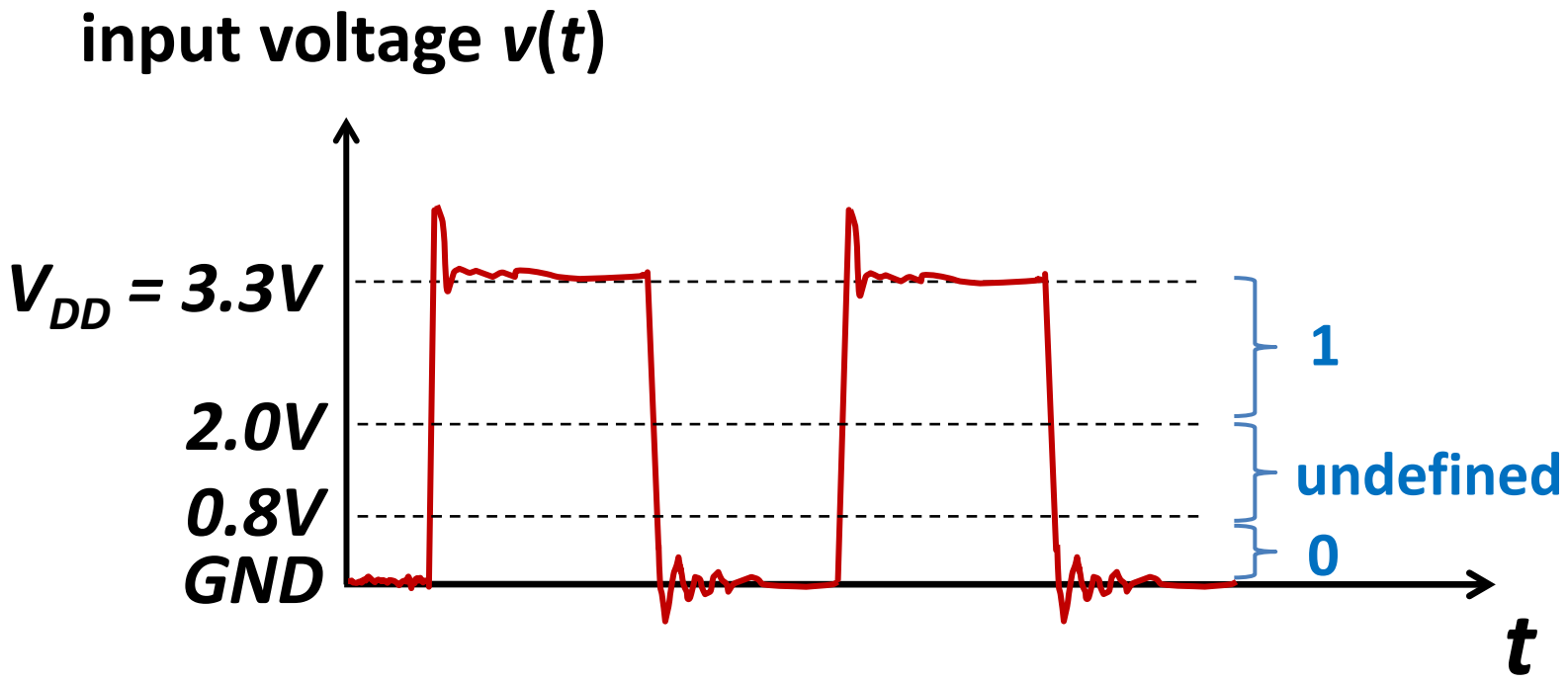
How many rows are in a truth table for:

n=4? *n*=5? *n*=8?

However, there is a practical limit to the number of inputs because of the electronic characteristics of the transistors. This is called "***fan-in***".

©2026 Dr. D. Capson. All rights reserved.

What do real "digital" signals look like?



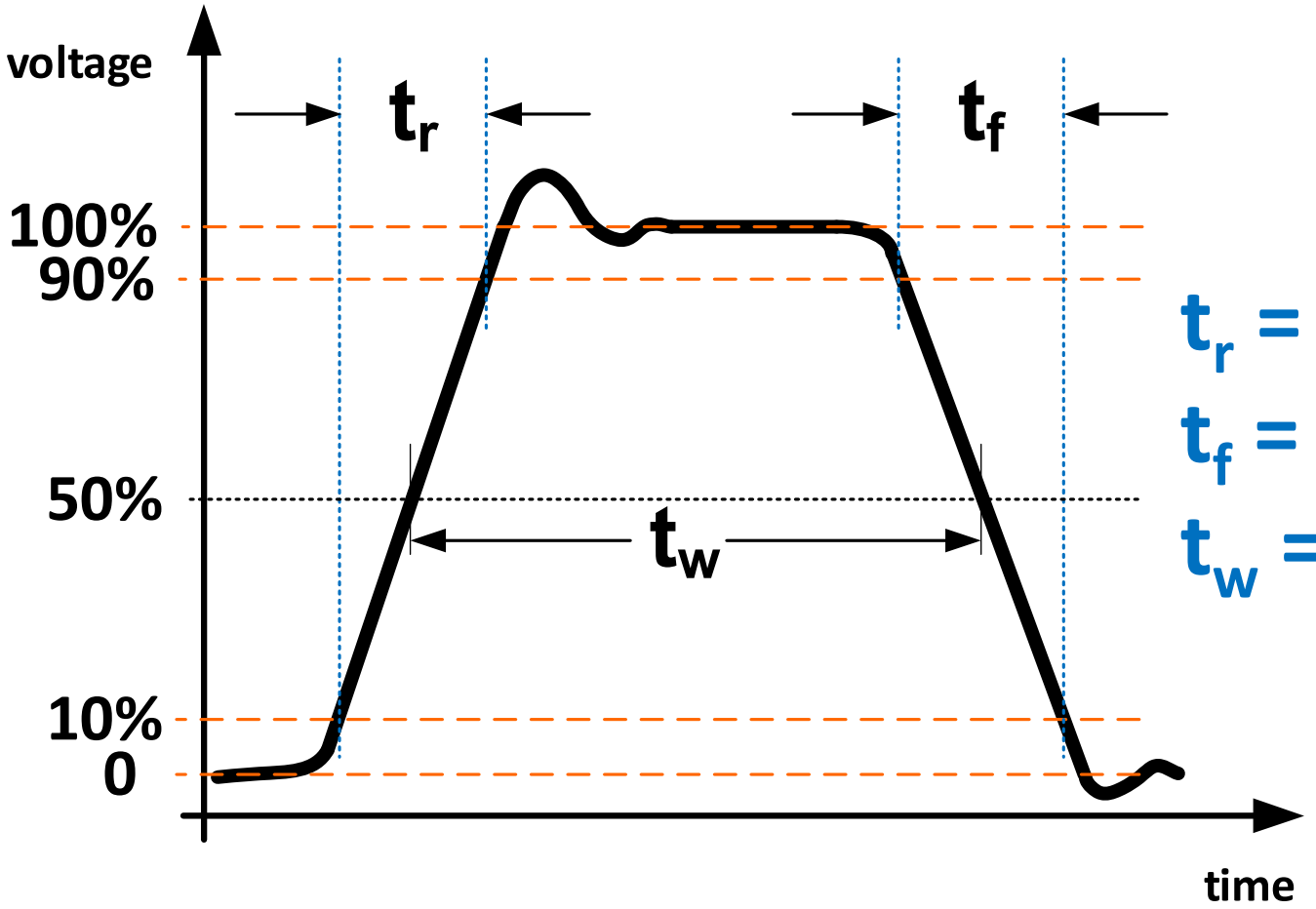
Notes:

(1) voltage ranges are different for inputs and outputs to provide noise "margin"

(2) other common values used for V_{DD} are 5V, 2.8V, 1.8V

©2026 Dr. D. Capson. All rights reserved.

Basic measurements of real “digital” signals



t_r = rise time

t_f = fall time

t_w = pulse width

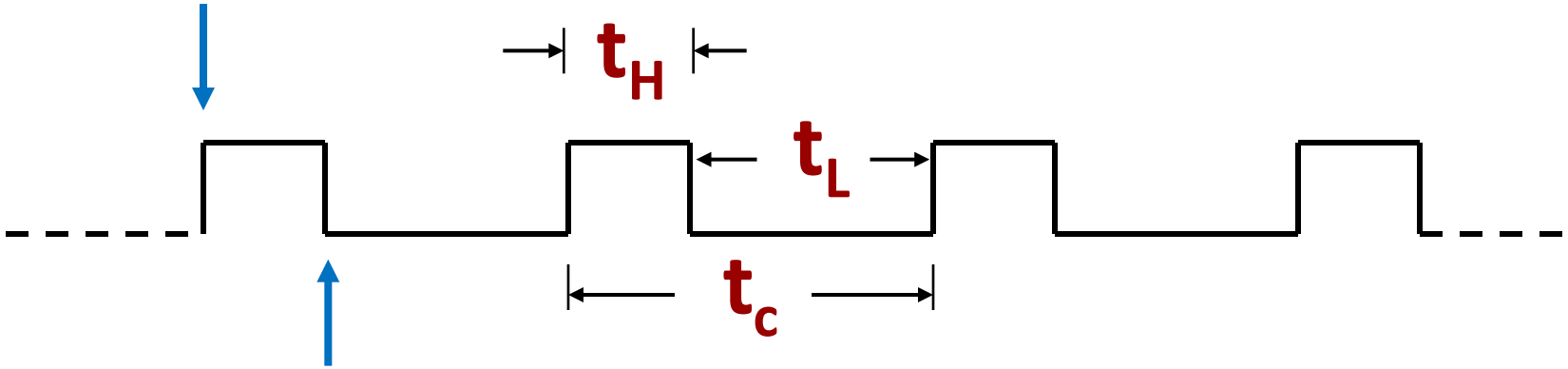
Note:

Sometimes t_w is measured at the switching threshold voltage which is usually <50%. This leads to a larger value for t_w

©2026 Dr. D. Capson. All rights reserved.

Frequency and Duty Cycle

leading/rising/positive edge



trailing/falling/negative edge

frequency (Hz) is the inverse of period (time):

$$f_c = 1/(t_H + t_L) = 1/t_c$$

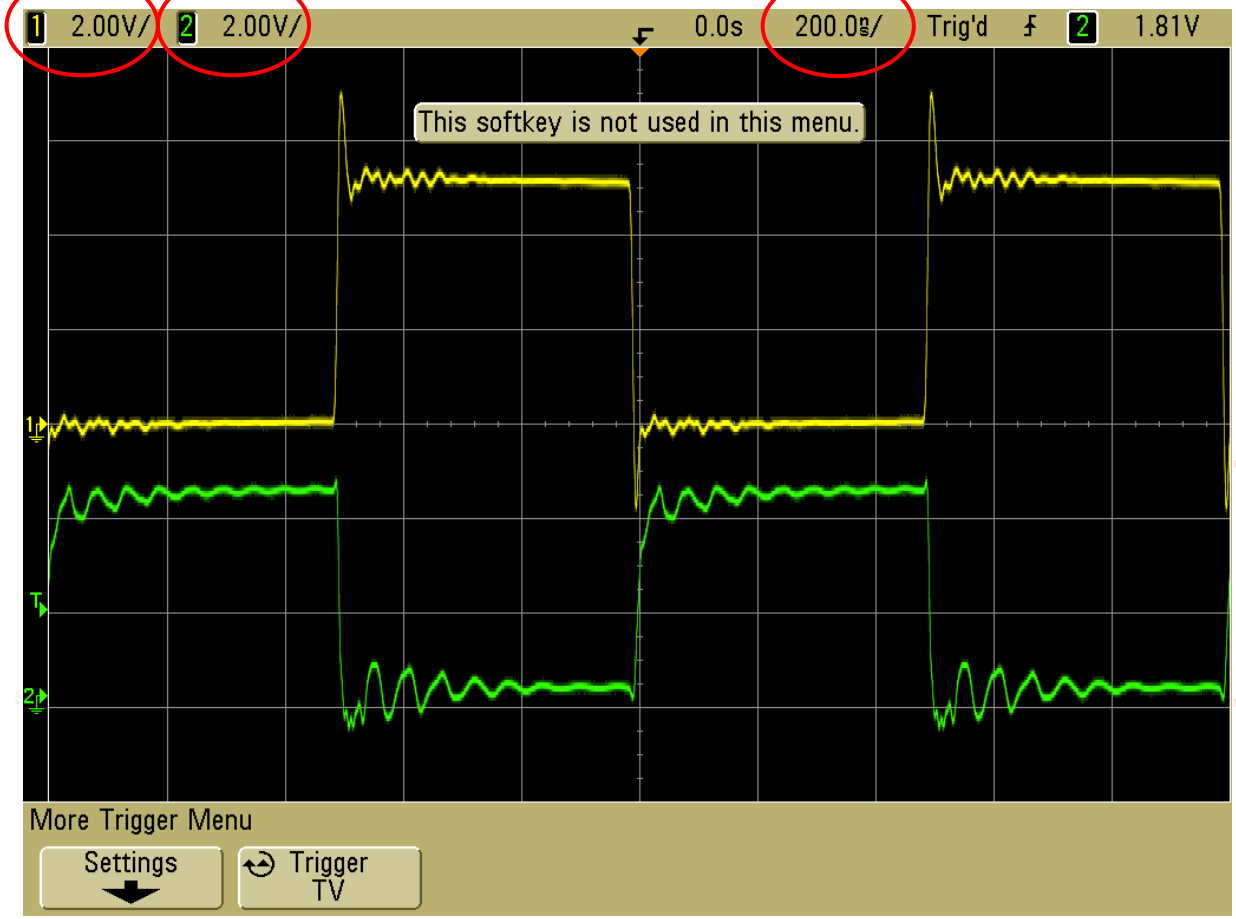
duty cycle is defined as the ratio: t_H / t_C

©2026 Dr. D. Capson. All rights reserved.

Display of real signals on Oscilloscope

Vertical sensitivities: *2 V/division*

Horizontal sensitivity: *200 ns/division*

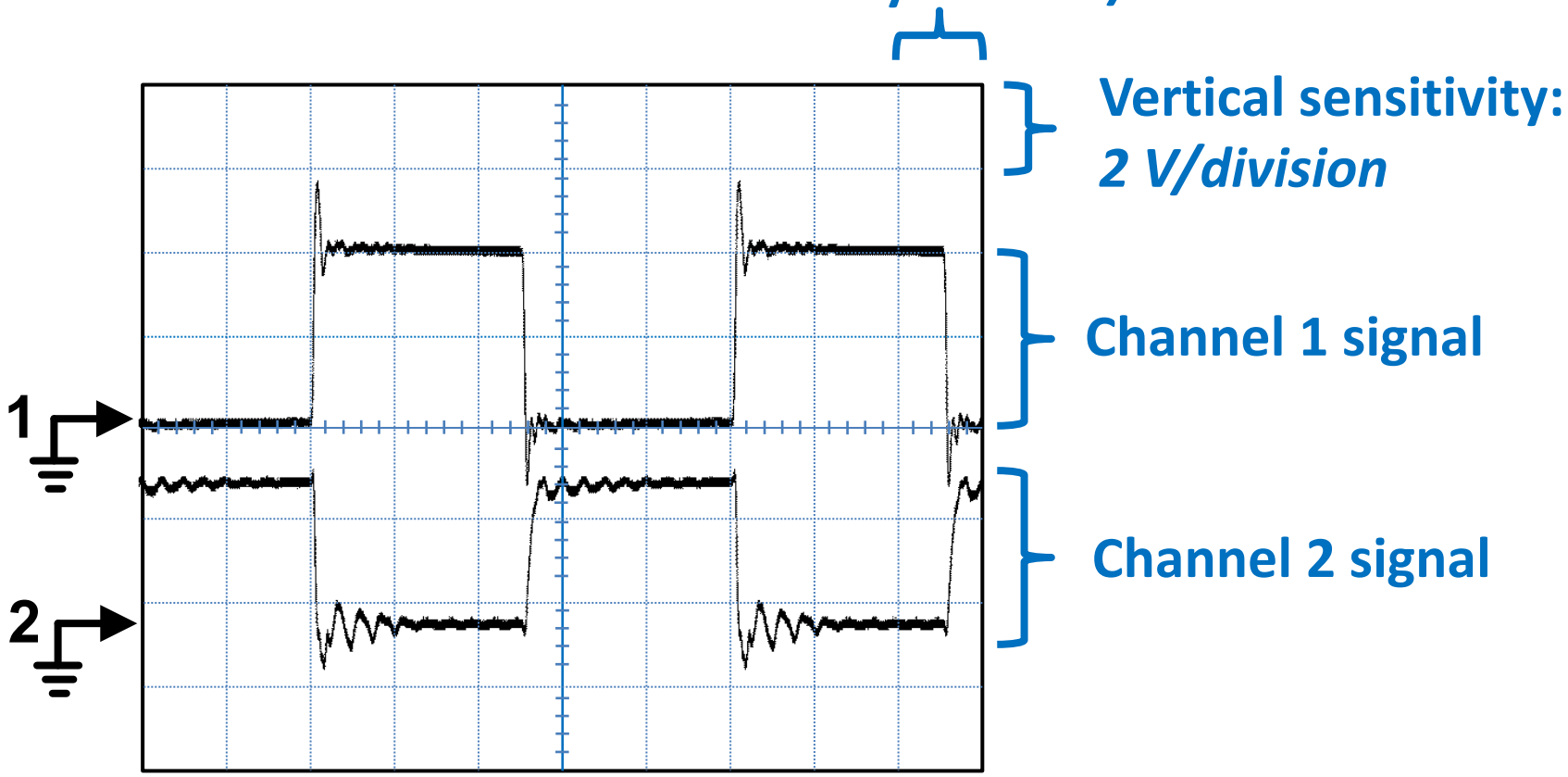


Channel 1 **(e.g. input to a NOT gate)**

Channel 2 **(e.g. output of the NOT gate)**

©2026 Dr. D. Capson. All rights reserved.

Display of real signals on ‘scope



Given horizontal & vertical sensitivities as shown above, what is the frequency, period, and duty cycle of the waveforms ? What are the logic level voltages ?

©2026 Dr. D. Capson. All rights reserved.

Examples:

1 MHz → 1 μs period = 1,000 ns = 1,000,000 ps

10 MHz → 0.1 μs period = 100 ns = 100,000 ps

100 MHz → 10 ns period = 10,000 ps

1 GHz → 1 ns period = 1,000 ps

2 GHz → 0.5 ns period = 500 ps

3 GHz → 0.333 ns period = 333 ps

5 GHZ → 0.200 ns period = 200 ps

Speed of light is approximately:

3.00×10^8 m/s.

How far does light travel in 1 ns?

About 30 cm !!

©2026 Dr. D. Capson. All rights reserved.